UM-1 provenance: where every ingredient of the universe model comes from

dirac16complex project (with Claude Opus 5.5)

Contents

1. The order **2**

2. Ingredients taken from the Revision record (TAKEN) **2**

3. Standard physics used as input (STANDARD) **3**

4. Ingredients new in this model (ASSUMED or INVENTED in `MODEL.md`) **5**

5. Computations and tools **5**

6. Status **6**

Version 1.1 of 2026-10-17, for `MODEL.md` (the model), `FIELD_EQUATIONS.md` (its field equations) and `SOLUTION.md` (an approximate solution). Repository `https://github.com/once-ere/Dirac_claude`, folder `Revision/universe_model/`; the model was started on top of commit `13f9fb9`.

1. The order

The author's order of 2026-10-17 (summarised; the quoted words are the author's): create "a new model of the Universe" that includes Einstein-Lovelock gravity and "the dirac16complex quantized spinor field (which is hypothesied to represent dark matter)", coupled to gravity "using a non-trivial canonical spin connection", with "new CRITICAL interactions" of these fields with photons, neutrinos, electrons, up and down quarks and the W and Z bosons, "do NOT include the Higgs particle"; able to model the CMB, the CNB and BAO, so that the metric must be "very general, but perhaps expressible as a power series of terms of decreasing magnitude"; new phenomenological interactions "consistent with qcd and electro-weak models"; and, CRITICALLY, mass is to be given to all non-massless particles (all except the graviton, the photon "and perhaps the quanta of the dirac16complex field") "by appropriate wave function dependence on the space coordinate x0 (either explicit, derived, or assumed), which must be periodic in x0 with period $\pi/(6H)$", with the uncertainty relation $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$ as the reason. Then: push the model with provenance (.md and .pdf), compute the coupled field equations, compute an accurate approximate solution, document each (.md and .pdf), and complete the open stages.

The mass-from-x_0 idea and the periodicity requirement are the AUTHOR's. The assembly of the model, the choices labelled ASSUMED and INVENTED in `MODEL.md`, the derivations and the numerics are by Claude Opus 5.5 working in this repository.

2. Ingredients taken from the Revision record (TAKEN)

ingredient	**source in this repository**	**status there**
the author's coordinates $x_0, \ldots, x_7$ and the corrected primordial metric (version 2)	`Revision/SPEC.md` section 1; `Revision/metric_warped/`; `Revision/docs/METRIC_SELF_CONSISTENCY.md`	sympy 14/14, Wolfram 13/13
the Einstein-Lovelock equations with a general diagonal source, the author's weights $w = (1, 1/32, 1/1152)$, $\Lambda = -10H^2$	`Revision/field_equations_a4/a4-equations.json` (keys generalSource, einstein)	Wolfram 57/57, sympy 69/69
the Lovelock tensors of the corrected metric	`Revision/gkd_lovelock/results/lovelock-tensors.json`	W4
the dirac16complex Lagrangian and the γ matrices of $Cl(4,4)$ in the author's order	`Revision/SPEC.md` section 3; `Revision/algebra/gammas.json`	exact

the canonical spin connection, $\gamma^M\Omega_M = \frac{3}{2}He^{a_4}b_4^{-1/2}\gamma^{x_0} + (u - \frac{1}{2}\dot{a}_4)\gamma^{x_4}$	`Revision/lead_checks/emt_divergence_and_spin_connection.py`	10/10
the reduction to 2×2 blocks, $h_j = j[-(i/f)\sigma_1\partial_y + M\sigma_2 + \kappa k\sigma_3] + v$, the fold and tip conditions, the exact $k = 0$ spectra	`Revision/kohn_sham/ks-theory.json`	Wolfram 51/51, sympy 61/61
the fold-band quanta $\varepsilon = e^{-a_4}\|k\|$ (massless), the bulk band $\varepsilon \geq m$	`Revision/kohn_sham/results/`, `Revision/kohn_sham/tip_convergence/`	45/45, 7/7
the condensate law $S = S_0 e^{a_4} b_4^{-2}$	`Revision/field_equations_a4/a4-equations.json` (fields)	exact
theorem T3 (the map $\chi \to \sigma_2\chi$, $M \to -M$, $\theta \to \pi - \theta$)	`Revision/pairing/kohn_sham/t3-theory.json`	exact
the dark-sector findings for the corrected metric (free gas $w_{eff} = 1/3$, condensate $w_{eff} = \frac{5}{9}w_c$, bulk band)	drafts of phase W6, `handoff/drafts_2026-10-17/w6/`	drafts (37/37, 13/13, 10/10), not yet the record

The Kohn-Sham cross-check of the record was NOT complete when this model was started: `HANDOFF.md` 0.10, "W5 STATE AT THE STOP OF 2026-10-17". Nothing of the model depends on the three failing comparisons, which concern numbers at the 10^{-10} level.

3. Standard physics used as input (STANDARD)

input	**value used**	**source**
electron mass	$m_e = 0.51099895$ MeV	Particle Data Group (from memory of the PDG tables; to be checked against the current edition)
up, down quark masses ($\overline{MS}$, 2 GeV)	$m_u = 2.16$ MeV, $m_d = 4.70$ MeV	PDG (as above)
W and Z masses	$m_W = 80.3692$ GeV, $m_Z = 91.1880$ GeV	PDG 2024 (as above)
the electroweak ρ parameter	$\rho_0 = 1.00038 \pm 0.00020$	PDG electroweak review (as above); the target of the m_{KK} fit of `SOLUTION.md`
the neutrino mass of the one generation	$m_\nu = 0.05$ eV	chosen at $\sqrt{\|\Delta m^2_{31}\|}$ (ASSUMED)

electroweak scale, fine-structure constant	$v = 246.22$ GeV, $\alpha^{-1} = 137.036$	PDG (as above)
neutrino mass splittings	$\Delta m^2_{21} = 7.5 \times 10^{-5}$ eV^2, $\lvert\Delta m^2_{31}\rvert = 2.5 \times 10^{-3}$ eV^2	PDG (as above)
CMB temperature	$T_0 = 2.7255$ K	Fixsen 2009 (as above)
cosmological parameters	$\Omega_b h^2 = 0.02237$, $\Omega_c h^2 = 0.1200$, $H_0 = 67.36$ km/s/Mpc, $N_{eff} = 3.044$, $z_* = 1089.92$, $z_{drag} = 1059.94$, $r_{drag} = 147.09$ Mpc, $100\,\theta_* = 1.04110$; with N_{eff} free: $N_{eff} = 2.99 \pm 0.17$	Planck 2018 results VI, Table 2 (as above); θ_* is the target of the h fit, the rest COMPARISON
BAO	the 12 BAO-only consensus measurements of eBOSS DR16, $z = 0.15$ to 2.33	Alam et al. 2021 (PRD 103, 083533); the table of `https://www.sdss4.org/science/final-bao-and-rsd-measurements/`, read on 2026-10-17
fitting formulae	z_* (Hu and Sugiyama 1996), z_{drag} (Eisenstein and Hu 1998), the first-peak shift ϕ_1 (Hu, Fukugita, Zaldarriaga and Tegmark 2001), $N_{eff} = 3.044$ for three species (Bennett et al. 2021)	literature (from memory), calibrated on Planck ΛCDM in `solution/cosmology.py`
QCD: confinement of u, d into p, n; nucleon masses	$m_p = 938.272$ MeV, $m_n = 939.565$ MeV	PDG (as above)
the electroweak chiral Lagrangian (Higgsless symmetry breaking) and boundary-condition symmetry breaking in extra dimensions	textbook and review literature on Higgsless models and on fermion localisation in extra dimensions	concepts only; no numbers

All standard numbers are quoted from memory, except the eBOSS DR16 table (read from the SDSS page above); they are labelled as such in the outputs and are used only as targets of fits and as comparisons, never as derived results.

4. Ingredients new in this model (ASSUMED or INVENTED in `MODEL.md`)

ingredient	label	where
the metric as a power series around the corrected primordial metric, with the bookkeeping parameter $\epsilon \sim 10^{-5}$	ASSUMED	`MODEL.md` section 3
the Standard-Model fields as eight-dimensional fields of $SU(3)_c \times SU(2)_L \times U(1)_Y$ on the cell, including the gluons	ASSUMED	section 4.3
the chiral zero modes from the bulk masses μ_ψ and the fold conditions θ_ψ	INVENTED (built on the record's fold conditions)	section 4.3
no Higgs field; the tip sigma field Σ (non-linear, no radial mode) and its couplings v_Σ, λ_f	INVENTED	section 4.3
the mass mechanism: four-dimensional masses as eigenvalues of the x_0 operators on the periodic cell, with the uncertainty bound	the AUTHOR's idea, made precise here	section 4.4
the portal operators P1 to P5	INVENTED	section 4.5
the requirement of slow deflation after the primordial era	DERIVED requirement (mass constancy)	section 4.4, `SOLUTION.md`
(1.1) the cubic Lovelock weight $w_3 = 1/768$ and $\Lambda = -24H^2 + \delta\Lambda$	CHANGED (forced by the solution: the author's weights give repulsive gravity in the only static vacuum)	section 4.1, `SOLUTION.md` section 3
(1.1) the block stabiliser	ASSUMED (its origin OPEN)	section 4.1
(1.1) the fold-reflection parity that keeps the odd-parity Ψ dark matter stable	ASSUMED	section 8, `SOLUTION.md` section 5

5. Computations and tools

- Symbolic: Python 3.14.5, sympy 1.14.0 (`field_equations/derive_field_equations.py`).
- Numerical: numpy 2.4.6, no scipy; matplotlib 3.11.0 for the figures (`solution/`). The Kohn-Sham solution uses the record's reference solver `Revision/kohn_sham/reference/ks_fd.py` and its coefficients (`run_reference.py`) UNCHANGED (TAKEN), with the portal potentials added by a wrapper (`solution/ks_coupled.py`).
- Documents: the Markdown files are the sources; `tools/md2tex.py` converts each to LaTeX

and `tools/build_pdfs.py` compiles it with pdflatex (MiKTeX 26.5), so the .md and the .pdf say the same thing.

- Every calculation runs well within the limit of 423 s per calculation of this project.

6. Status

deliverable	**file**	**status**
the model	`MODEL.md`, `MODEL.pdf`	version 1.1
this provenance	`PROVENANCE.md`, `PROVENANCE.pdf`	version 1.1
the coupled field equations	`FIELD_EQUATIONS.md`, `.pdf`; `field_equations/`	see that document
an approximate solution	`SOLUTION.md`, `.pdf`; `solution/`	version 1 (checks in `solution/reports/solution-checks.json`)